

## Project Synopsis: LPU Exam Cracker

**Project Category:** AI-Powered Educational Tool / Cloud-Native Application

**Developer:** Harshnath, Vishwajeet Singh

**Target Audience:** Students of Lovely Professional University (LPU)

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### 1. Introduction

The **LPU Exam Cracker** is an AI-driven web application designed to streamline exam preparation for university students. By leveraging Generative AI and cloud-native services, the platform transforms dense syllabus PDFs into actionable study kits, including concise summaries, weightage-based topic highlights, and practice assessments.

### 2. Problem Statement

LPU students often face "information overload" due to vast syllabi across multiple subjects.

- **Time Constraints:** Manually summarizing units before Mid-Term (MTE) or End-Term (ETE) exams is time-consuming.
- **Pattern Recognition:** Students struggle to identify "High-Priority" topics from raw syllabus documents.
- **Assessment Gap:** There is a lack of instant, syllabus-specific practice questions (MCQs) for self-evaluation.

### 3. Objectives

- **Automated Summarization:** Generate a 5-point "Must-Know" summary for every unit in the syllabus.
  - **Weightage Analysis:** Use AI to predict high-weightage topics based on LPU's standard exam patterns.
  - **Interactive Testing:** Generate context-aware MCQs to facilitate active recall.
  - **Mobile-First Access:** Deploy as a Progressive Web App (PWA) for instant access via smartphone browsers.
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### 4. Proposed Solution & Features

The application allows a student to upload an official LPU syllabus PDF. The system then processes the text through a Large Language Model (LLM) to extract structured academic data.

**Key Features:**

- **Smart Dropzone:** Drag-and-drop interface for PDF uploads.
- **AI Professor Module:** A specialized prompt-engineered engine that understands LPU-specific academic terminology (CA, Units, ETE).
- **Study Dashboard:** A clean UI to view summaries, priority topics, and take timed quizzes.

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**5. Technical Architecture (AWS Stack)**

To ensure scalability and leverage DevOps best practices, the project utilizes an **AWS Serverless Architecture**.

| Component     | Technology                    | Role                                               |
|---------------|-------------------------------|----------------------------------------------------|
| Frontend      | Next.js 15 + Tailwind CSS     | Responsive UI and LPU-themed branding.             |
| Hosting       | AWS Amplify                   | Continuous Deployment (CI/CD) and hosting.         |
| Storage       | Amazon S3                     | Secure storage for uploaded syllabus PDFs.         |
| AI Engine     | Amazon Bedrock (Claude/Llama) | Processing PDF text for summaries and MCQs.        |
| Backend Logic | AWS Lambda                    | Trigger-based serverless functions for AI calls.   |
| Database      | Amazon DynamoDB               | Storing processed summaries for instant retrieval. |

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**6. Future Scope**

- **PYQ Integration:** Allowing students to upload Previous Year Questions (PYQs) for even more accurate weightage prediction.
  - **Collaborative Study:** Shared "Study Rooms" where students can see summaries generated by peers for common courses.
  - **Voice Summaries:** Implementing Text-to-Speech (TTS) to allow students to listen to unit summaries on the go.
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